Arithmetic series

1

a $S_{10} = 195$ **b** $S_{10} = 0$ **c** $S_{10} = 280$ **d** $S_{10} = -310$

2

a $S_{15} = 450$ **b** $S_{15} = -45$ **c** $S_{15} = 105$

3

a $S_{10} = 47.5$ **b** $S_{10} = -53.75$ **c** $S_{10} = 85$ **d** $S_{10} = 5$

4 $S_{20} = 210\pi$

5

a $d = -8$ **b** $S_{14} = -798$

6 $S_{14} = \dfrac{273}{2}$

7

a $n = 18$ **b** $S_{18} = 450$

8 The sum of an arithmetic series is calculated as

$$S_n = \frac{n}{2}\left(2a_1 + (n-1)d\right)$$

where n is the number of terms, a_1 is the first term and d is the difference between consecutive values.
Substitute $a_1 = 1$ and $d = 2$, we have:

$$\begin{aligned} S_n &= \frac{n}{2}\left(2 \times 1 + (n-1) \times 2\right) \\ &= \frac{n}{2}\left(2 + 2n - 2\right) \\ &= \frac{n}{2}\left(2n\right) \\ &= n^2 \end{aligned}$$

9

a $S_{40} = 4920$ **b** $S_{40} = -2260$

10 $n = 10$

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11 180

12 $\sum_{k=1}^{11} \left(-\frac{1}{2}\right)^{k-1}$

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